Example

Problem

- Calculate the total number of iterations for the given **for** loop structure, i.e., how many times Line 4 executed?

```
Algorithm Loop (n):
L1. s = 0
L2. For i = 1 to 2n do
L3.         For j = i to 2n do
L4.                 s = s + i
```

Outer loop (k-th)	Innter loop (times)
1	2n
2	2n-1
3	2n-2
…	…
2n	1

$$\sum_{i=1}^{2n} i = \frac{2n(2n+1)}{2} = 2n^2 + n$$